

For the multivariable normal law, we can compute the mean

$$\mathcal{E}\{\mathbf{x}\} = \boldsymbol{\mu}_x,$$

and the covariance matrix

$$\mathcal{E}\{(\mathbf{x} - \boldsymbol{\mu}_x)(\mathbf{x} - \boldsymbol{\mu}_x)^T\} = \mathbf{R}_x.$$

Like the scalar normal density, the multivariable law is described entirely by the two parameters $\boldsymbol{\mu}$ and $\mathbf{R}$, the difference being that the multivariable case is described by matrix parameters rather than scalar parameters. In Eq. (D.19) we require the inverse of $\mathbf{R}_x$ and have thus implicitly assumed that this covariance matrix is nonsingular. [See Parzen (1962) for a discussion of the case when $\mathbf{R}_x$ is singular.]

D.4 Stochastic Processes

In a study of dynamic systems, it is natural to have random variables that evolve in time much as the states and control inputs evolve. However, with random time variables it is not possible to compute z-transforms in the usual way; and furthermore, because specific values of the variables have little value, we need formulas to describe how the means and covariances evolve in time. A random variable that evolves in time is called a **stochastic process**, and here we consider only discrete time.

Suppose we deal first with a stochastic process $w(k)$, where w is a scalar distributed according to the density $f_w(N)$. Note that the density function depends on the time of occurrence of the random variable. If a variable has statistical properties (such as f_w) that are independent of the origin of time, then we say the process is **stationary**. Considering values of the process at distinct times, we have separate random variables, and we define the covariance of the process w as

$$R_w(j, k) = \mathcal{E}(w(j) - \bar{w}(j))(w(k) - \bar{w}(k)). \quad (\text{D.20})$$

If the process is stationary, then the covariance in Eq. (D.20) depends only on the magnitude of the difference in observation times, $k - j$, and we often will write $R_w(j, k) = R_w(k - j)$ and drop the second argument. Because a stochastic process is both random and time dependent, we can imagine averages that are computed over the time variable as well as the averages defined in Section D.2. For example, for a stationary process $w(n)$ we can define the mean as

$$\tilde{w}(k) = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N w(n+k), \quad (\text{D.21})$$

and the second-order mean or autocorrelation

$$\mathcal{E}\{(w(j) - \tilde{w})(w(k) - \tilde{w}(k))\} = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N \{(w(n+j) - \tilde{w}(j))(w(n+k) - \tilde{w}(k))\}. \quad (\text{D.22})$$

For a stationary process, the time average in Eq. (D.21) is usually equal to the distribution average, and likewise the second-order average in Eq. (D.22) is the same as the covariance in Eq. (D.20). Processes for which time averages give the same limits as distribution averages are called **ergodic**.

A very useful aid to understanding the properties of stationary stochastic processes is found by considering the response of a linear stationary system to a stationary input process. Suppose we let the input be w , a stationary scalar process with zero mean and covariance $R_w(j)$, and suppose we take the output to be $y(k)$. We let the unit-pulse response from w to y be $h(j)$. Thus from standard analysis (see Chapter 4), we have

$$y(j) = \sum_{k=-\infty}^{\infty} h(k)w(j-k), \quad (\text{D.23})$$

and the covariance of $y(j)$ with $y(j+\ell)$ is

$$\begin{aligned} R_y(\ell) &= \mathcal{E}y(j+\ell)y(j) \\ &= \mathcal{E} \left\{ \sum_{k=-\infty}^{\infty} h(k)w(j+\ell-k) \right\} \left\{ \sum_{n=-\infty}^{\infty} h(n)w(j-n) \right\}. \end{aligned} \quad (\text{D.24})$$

Because the system unit-pulse response, $h(k)$, is not random, both $h(k)$ and $h(n)$ can be removed from the integral implied by the $\mathcal{E}$ operation, with the result

$$R_y(\ell) = \sum_{k=-\infty}^{\infty} h(k) \sum_{n=-\infty}^{\infty} h(n) \mathcal{E}\{w(j+\ell-k)w(j-n)\}. \quad (\text{D.25})$$

The expectation in Eq. (D.25) is now recognized as $R_w(\ell-k+n)$, and substituting this expression in Eq. (D.25), we find

$$R_y(\ell) = \sum_{k=-\infty}^{\infty} h(k) \sum_{n=-\infty}^{\infty} h(n) R_w(\ell-k+n). \quad (\text{D.26})$$

Equation (D.26) is not especially enlightening, but the z -transform of it is. We proceed with several simple steps as follows

$$\begin{aligned} z\{R_y(\ell)\} &= \sum_{\ell=-\infty}^{\infty} R_y(\ell)z^{-\ell} \\ &= \sum_{\ell=-\infty}^{\infty} \sum_{k=-\infty}^{\infty} h(k) \sum_{n=-\infty}^{\infty} h(n) R_w(\ell-k+n) z^{-\ell}. \end{aligned}$$

Exchanging the order, because $h(k)$ and $h(n)$ do not depend on ℓ , we have

$$z\{R_y(\ell)\} = \sum_{k=-\infty}^{\infty} h(k) \sum_{n=-\infty}^{\infty} h(n) \sum_{\ell=-\infty}^{\infty} R_w(\ell - k + n) z^{-\ell}.$$

Now we let $m = \ell - k + n$ in the last sum, leading to

$$z\{R_y(\ell)\} = \sum_{k=-\infty}^{\infty} h(k) \sum_{n=-\infty}^{\infty} h(n) \sum_{m=-\infty}^{\infty} R_w(m) z^{-(m+k-n)}.$$

Finally we use the fact that $z^{-(m+k-n)} = z^{-m} z^{-k} z^n$ and distribute these terms to the corresponding sums with the result

$$\begin{aligned} z\{R_y(\ell)\} &= \sum_{k=-\infty}^{\infty} h(k) z^{-k} \sum_{n=-\infty}^{\infty} h(n) z^n \sum_{m=-\infty}^{\infty} R_w(m) z^{-m} \\ &= S_y(z). \end{aligned} \quad (\text{D.27})$$

For reasons soon to be clear, we call the z -transform of R_y the **spectrum of y** and use the symbol $S_y(z)$, and similarly for w and S_w . With these symbols and recognition that the z -transform of the unit-pulse response is the system-transfer function, $H(z)$, Eq. (D.27) becomes

$$S_y(z) = H(z) H(z^{-1}) S_w(z). \quad (\text{D.28})$$

To give an interpretation of Eq. (D.28) we make two observations. First note that $R_y(0) = \mathcal{E}(y^2)$ is the mean-square value or power in the y -process. By the inverse transform integral, we have

$$\begin{aligned} \overline{y^2} &= R_y(0) \\ &= \frac{1}{2\pi j} \oint S_y(z) \frac{dz}{z} \\ &= \frac{1}{2\pi j} \oint H(z) H(z^{-1}) S_w(z) \frac{dz}{z}. \end{aligned} \quad (\text{D.29})$$

Now, as a second step, we suppose that $H(z)$ is the transfer function of a very narrow bandpass filter centered at ω_0 , so that $H(z) H(z^{-1})$ is $|H(e^{j\omega_0 T})|^2$ and is nearly zero except at ω_0 . Then the integral in Eq. (D.29) can be approximated by assuming that $S_w(z)$ is nearly constant at the value $S_w(e^{j\omega_0 T})$, where $|H|^2$ is nonzero and can thus be removed from the integral. The result is

$$\begin{aligned} \overline{y^2} &= S_w(e^{j\omega_0 T}) \frac{1}{2\pi j} \oint H(z) H(z^{-1}) \frac{dz}{z} \\ &= S_w(e^{j\omega_0 T}) K. \end{aligned} \quad (\text{D.30})$$

In Eq. (D.30) we have defined the integral as a constant dependent on the exact area of the narrow-band characteristic of $H(z)$. But now we can give good intuitive meaning to Eq. (D.30). The mean square of the output of a very-narrow-band filter is proportional to the S_w . If S_w is constant for all z , we say the process is white

(after the spectrum of white light, which has equal intensity at all frequencies). Hence we call S_w **the power spectral density** of the w -process. Equation (D.28) is the fundamental formula for transform analysis of linear constant systems with stochastic inputs.

An alternative to transform analysis and Eq. (D.28) is transient analysis via the state-variable formulation, described in Section 4.3.3. In this case, we take the system equations to be

$$\begin{aligned} \mathbf{x}(k+1) &= \Phi \mathbf{x}(k) + \Gamma_1 \mathbf{w}(k), \\ \mathbf{y}(k) &= \mathbf{H} \mathbf{x}(k). \end{aligned} \quad (\text{D.31})$$

We assume that the system starts at $k = 0$ with the initial value

$$\begin{aligned} \mathcal{E}\{\mathbf{x}(0)\} &= \mathbf{0}, \\ \mathcal{E}\{\mathbf{x}(0)\mathbf{x}^T(0)\} &= \mathbf{R}_{xx}(0; 0) \\ &= \mathbf{P}_x(0), \end{aligned}$$

and that $\mathbf{w}(k)$ is a stationary process with covariance

$$\begin{aligned} \mathcal{E}\{\mathbf{w}(k)\mathbf{w}^T(k+j)\} &= \mathbf{0} \quad (j \neq 0) \\ &= \mathbf{R}_w \quad (j = 0). \end{aligned} \quad (\text{D.32})$$

Note that by Eq. (D.27), $S_w(z) = \mathbf{R}_w$, a constant, and hence $\mathbf{w}$ is a white process. With these conditions, we can compute the evolution of $\mathbf{P}_x$, the autocovariance of the state at time k . Thus

$$\begin{aligned} \mathcal{E}\{\mathbf{x}(k)\mathbf{x}^T(k)\} &= \mathbf{R}_{xx}(k; k) \\ &= \mathbf{P}_x(k), \end{aligned}$$

and

$$\begin{aligned} \mathcal{E}\{\mathbf{x}(k+1)\mathbf{x}^T(k+1)\} &= \mathbf{P}_x(k+1) \\ &= \mathcal{E}\{(\Phi \mathbf{x}(k) + \Gamma_1 \mathbf{w}(k))(\Phi \mathbf{x}(k) + \Gamma_1 \mathbf{w}(k))^T\} \\ &= \Phi \mathcal{E}\{\mathbf{x}(k)\mathbf{x}^T(k)\} \Phi^T + \Phi \mathcal{E}\{\mathbf{x}(k)\mathbf{w}^T(k)\} \Gamma_1^T \\ &\quad + \Gamma_1 \mathcal{E}\{\mathbf{w}(k)\mathbf{x}^T(k)\} \Phi^T + \Gamma_1 \mathcal{E}\{\mathbf{w}(k)\mathbf{w}^T(k)\} \Gamma_1^T. \end{aligned} \quad (\text{D.33})$$

Because the center two terms in Eq. (D.33) are zero by Eq. (D.32)⁵ we reduce Eq. (D.33) to

$$\begin{aligned} \mathbf{P}_x(k+1) &= \Phi \mathbf{P}_x(k) \Phi^T + \Gamma_1 \mathbf{R}_w \Gamma_1^T, \\ \mathbf{P}_x(0) &= \text{a given matrix initial condition.} \end{aligned} \quad (\text{D.34})$$

Equation (D.34) is the fundamental equation for the time-domain analysis of discrete systems with stochastic inputs. Note that Eq. (D.34) represents a non-stationary situation because the covariance of $\mathbf{x}(k)$ depends on the time k of

⁵ $\mathbf{x}(k)$ is a combination of $w(0), w(1), \dots, w(k-1)$, all of which are uncorrelated with $w(k)$.

the occurrence of $\mathbf{x}$. However, if all the characteristic roots of Φ are inside the unit circle, then the effects of the initial condition, $\mathbf{P}_x(0)$, gradually diminish, and $\mathbf{P}_x$ approaches a stationary value. This value is given by the solution to the (Lyapunov) equation

$$\mathbf{P}_x = \Phi \mathbf{P}_x \Phi^T + \Gamma_1 \mathbf{R}_w \Gamma_1^T. \quad (\text{D.35})$$

• E •

MATLAB Functions

Table E.1 MATLAB functions and pages where used

<i>Function (.m file)</i>	<i>Student Edition v5</i>	<i>MATLAB v 5.0</i>	<i>Control System Toolbox v 4.0</i>	<i>Digital Control Toolbox 3e</i>	<i>Page(s)</i>
acker	x		x		43,46,286,295,297
bode	x		x		33,38,134
c2d	x		x		99,107,116,281,288
connect			x		219
conv	x	x			28,63
d2c	x		x		470
damp	x		x		45,64
disrw				x	397,455
dlqr	x		x		377,409
dlyap	x		x		435,455
eig	x	x			45,109
feedback	x		x		29,62,288
impulse	x		x		18,131
initial			x		288
jdequiv				x	380
kalman	x		x		47,395,415
loglog	x	x			134
lqr	x		x		44
lqrd	x		x		380
lsim	x		x		15,49,326
lyap	x		x		455
margin	x		x		64
nyquist	x		x		242
place	x		x		43,46,286,295
plot	x	x			18
qss				x	431
qwc				x	429
refi				x	313
ripple				x	182,231
rlocfind	x		x		29
rlocus	x		x		29,216
roots	x	x			45
semilogx	x	x			134
set	x		x		181
ss	x		x		13,14,280
ssdata	x		x		108,288
step	x		x		15,18,29,62,131
tf	x		x		13,14,81
tfdata	x		x		99
tzero	x		x		110
zgrid			x		128
zpk	x		x		13,14,82

• F •

Differences Between MATLAB v5 and v4

All the MATLAB scripts in the text are based on MATLAB v5 and the Control System Toolbox v4, which we will refer to as “v5.” Some users of this text may be using MATLAB v4 with the Control System Toolbox v3 or earlier versions, which we will refer to as “v4.” This Appendix describes the differences between these two versions of MATLAB and is intended to guide the reader with v4 how to interpret the scripts in the text. As discussed in the Preface, the MATLAB descriptions in the text are not meant to be a replacement to the MATLAB manuals; rather, their intent is to lead the reader to the routine in MATLAB that can be used to carry out a specific calculation in the text. In some cases it may eliminate the need to look at the manual or help file, but in most cases it will primarily point the reader to the right routine.

The most significant difference is the way that systems are described; therefore, that topic will be discussed first in Section F.1 along with a table of some of the associated name changes of the routines. Another difference that especially affects users of this text is the way that systems are converted from continuous to discrete models, a topic covered in Section F.2. The changes in the calculation of optimal estimation parameters are discussed in Section F.3.

F.1 System Specification

In either version, a linear system is specified in one of three forms; transfer function, *tf*, zero-pole-gain, *zpk*, or state-space, *ss*. In v4, the *tf* description was specified by numerator and denominator polynomials, often referred to as *num* and *den* while in v5 both these quantities are appended in one system description variable, often called *sys*. As a result, the routines that require a transfer function system specification in v4 require *num* and *den* variables in the call sequence

rather than the one variable, such as `sys`. For example, in v4 the impulse and step functions described in Section 2.1.6 would be

```
impulse(num, den)    or
step(num, den).
```

Likewise, if a system is described by the zero-pole-gain form, three variables are required and the statements above would be

```
impulse(z, p, k)    or
step(z, p, k),
```

or, if in the state-space form, four matrices are required and the statements would be

```
impulse(A, B, C, D)    or
step(A, B, C, D).
```

In v5, the statements on page 18 are the same for any model form, that is

```
impulse(sys)    or
step(sys).
```

To convert from one form to another in v4, conversion routines are used. For example, to convert from ss to tf, one writes

```
[num,den] = ss2tf(A,B,C,D).
```

For other conversions, v4 has `ss2zp`, `tf2ss`, `tf2zp`, `zp2tf`, and `zp2ss` for the specific conversion required. None of these exist in v5; rather, all conversions are accomplished using `ss`, `tf`, and `zpk` as shown on page 14.

The v5 system description includes several properties in addition to the differential or difference equations parameters discussed above. The sample time, T_s , and the time delay, T_d , are two of the important new properties included within the system description. If $T_s = 0$, it means the system description is for a continuous set of *differential* equations. If $T_s \neq 0$, it means the system description is for a discrete set of *difference* equations. A nonzero value of T_d means that the system has a pure time delay. Because the system description includes information on whether the system is continuous or discrete, separate routines are usually not required for the two cases as they were for v4. For example, in v4 there is `dbode` for the discrete case and `bode` for the continuous case whereas in v5 there is only one `bode` for both cases. The routine itself determines from the system parameters whether it is continuous or discrete and then carries out the appropriate calculations. For example, on page 134 the text states that a bode plot can be obtained for a discrete system by using `bode`. In v4, it would require the

use of `dbode`. Table F.1 lists all of the routines mentioned in this text which have been combined into one version for v5.

A summary of how to adjust for these new features is:

- Anywhere in the text where there is a routine with an argument consisting of a system description (usually some variant of `sys`), use of that routine in v4 would require replacing that system description with the specific variables for the form of the system being used. For the `tf` form it could be `num, den`, for the `zpk` form it could be `z,p,k`, and for the `ss` form, it could be `A,B,C,D`.
- Separate versions of many routines that existed in v4 for the continuous and discrete cases have been eliminated in v5. Therefore, when performing calculations with discrete systems, it may be necessary to adjust to the discrete version of the routine from that stated in the text. The routines affected are listed in Table F.1.

Table F.1 Selected discrete MATLAB routines reduced to one version for v5

<i>MATLAB v4</i>	<i>MATLAB v5</i>
<code>bode, dbode</code>	<code>bode</code>
<code>damp, ddamp</code>	<code>damp</code>
<code>impulse, dimpulse</code>	<code>impulse</code>
<code>initial, dinitial</code>	<code>initial</code>
<code>lqe, dlqe</code>	<code>kalman</code>
<code>lsim, dlsim</code>	<code>lsim</code>
<code>nichols, dnichols</code>	<code>nichols</code>
<code>nyquist, dnyquist</code>	<code>nyquist</code>
<code>reg, dreg</code>	<code>reg</code>
<code>step, dstep</code>	<code>step</code>

F.2 Continuous to Discrete Conversion

Conversion of a continuous system to a discrete equivalent is first discussed in Section 4.3.1 on page 99 and comes up repeatedly throughout the text. This conversion was carried out using the `c2d` routine, which accomplishes the conversions for all cases in v5. To make matters somewhat confusing, there is also a `c2d` routine in v4, but with significantly reduced capabilities in that it would only accept systems in the state-space form and would only find the ZOH equivalent. Routines with different names exist in v4 to handle other forms, other equivalents, and time delay. All these capabilities were rolled into `c2d` for v5. The situation is summarized in Table F.2.

Table F.2 MATLAB discrete equivalent routines

<i>Calculation</i>	<i>MATLAB v4</i>	<i>MATLAB v5</i>
convert ss form using 'zoh' equivalent method	c2d	c2d
convert any form using any method	c2dm	c2d
convert ss form using 'zoh' method with input delay	c2dt	c2d with $T_d \neq 0$

F.3 Optimal Estimation

The calculation of the optimal steady state gain for an estimator in Eq. (9.97) is performed by MATLAB's `kalman` routine in v5. This routine also computes the complete optimal estimation system equations as part of its duties. In v4, the optimal gain is computed by `dlqe` and it is left for the user to construct the estimator equations. There are several versions of LQE calculations in v4 which have slightly different capabilities (`lqe`, `dlqe`, `lqew`, `dlqew`, `lqe2`, `lqed`). All of these have been replaced in v5 by `kalman` with the exception of `lqed` which was replaced by `kalmd`.

♦ References ♦

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